

Genomics

What is a Genome?

Genome is the complete set of genetic instruction that determine the traits of an organism. it determines how we would look, age, and grow. It will also determines how each of our cells, tissues, and organs work. it's made up of DNA, genes, and chromosomes :

- **DNA** is a molecule in the cells that carries the genetic information of an organism. Analogically it can be seen as the **code/building blocks** of an organism.
- **Genes** are segments of DNA that determine our traits. Human have around 20,000 to 25,000 genes, half of which we inherited from our mother and the other half from our father.
- **Chromosomes** are long, bundled strands of DNA, each which carries many genes. In total there're 2 sets of 23 chromosomes, each inherited from our biological parents.

Genome Sequence and Mapping

History of Genome Sequencing and Mapping

- The 1st successful study in **genomics** is in 1995, where a group of scientist manage to determine the DNA sequence of the entire genome of *Haemophilus influenzae*, a disease-causing bacterium.
- The Completion of **Human Genome Project** in 2003, with the human genome containing 21,000 genes that encode for 100,000 proteins.

Method of Genome Mapping

The set of techniques to sequence the entire genome of an organism is called **the whole genome shotgun method**.

Central Dogma

The central dogma for molecular biology is **an explanation of the flow of genetic information within a biological system**. it's often stated that "DNA makes RNA, and RNA makes Protein" by Francis Crick in 1957 and published in 1958.

3 Process of Central Dogma :

1. From existing DNA creating new DNA (Replication).
2. From DNA creating new RNA (Transcription).

3. From RNA creating new Proteins (Translation).

The process from DNA to RNA to Protein is called gene expression.

Replication

DNA replication is the process which the genome's DNA is copied in the cells. It's done because before a cell divides itself, it must replicate its genome so that each daughter cell ends up with the same genome. This can result in 1 cell replicating into millions or even trillions of cells.

Transcription

Transcription is the process which DNA is copied into mRNA, which carries the information needed for protein synthesis. The process of transcription is done in 2 steps which are:

1. A pre messenger mRNA is formed, with the involvement of RNA polymerase enzymes. This process relies on the **Watson-Crick base pairing**, with the result being a single strand of RNA, which is the reverse-complement of the original DNA sequence.
2. The pre messenger RNA is "edited" to produce the desired mRNA molecule in a process called **RNA Splicing**.

Nucleic Acid

Nucleic Acid is made up of 2 families of nitrogen bases, purines and pyrimidines, which are:

- Purines : 2 ring molecule. which are Adenine (A) & Guanine (G).
- Pyrimidines : Single ring molecule. which are Cytosine (C), Thymine (T -> Only in DNA), Uracil (U -> Only in RNA).

Replication

There's a few mechanisms for DNA replication, which are:

Semiconservative Replication

This mechanism uses each strand of DNA molecule to specify the synthesis for a new strand. Meselson and Stahl showed this by using heavy isotope of nitrogen and separating the replication products. This mechanism produces 2 molecules consisting of 1 new strand and 1 old strand.

Prokaryotic Replication

This Replication has a few characteristics:

- This replication starts at a single origin
- Unwinding the DNA requiring energy and causes torsional strain

- Semi discontinuous replication
- Synthesis occurring at the replication fork.

Eucaryotic Replication

This Replication have a few characteristics :

- Requiring multiple origins
- Enzymology of the replication is more complex
- Archaeal replication proteins is similar to eucaryotic proteins.
- Linear chromosomes have specialized ends.

DNA Mutation

DNA Mutation is the side-result of the all the process above, which can cause **Loss of gene expression control** and cause cancer.

DNA Repair

Main Point :

- Cell are constantly expose to DNA-damaging agents such as UV light.
- DNA repair restore damaged DNA
- Can be either specific or non specific

DNA Manipulation

Main Point :

- Restriction Enzymes cleave DNA at Specific times
- DNA ligase allows construction of recombinant molecules.
- Transformation allows introduction of foreign DNA to E. Coli.

Genomics Application : Molecular Cloning

Main Point :

- Host-Vector System allow propagation of foreign DNA in bacteria
- DNA libraries contain the entire genome of an organism
- Reverse Transcriptase can make DNA copy of RNA
- Hybridization allows identification of specific DNA in complex mixtures.
- Specific Clones can be isolated from a library